# North South University

Department of Electrical & Computer Engineering

# Project REPORT

**Course Code:** CSE215L

**Course Title:** Programming Language II

**Course Instructor**: Rifat Ahmed Hasan (RIH)

Project Name:

## SIMPLE GAMING CONSOLE

**Paper Due Date**: November 15th, 2023.

**Section**: 09

**Group Number**: 06

**Submitted To**: Md. Abdullah Al Sayed.

| Submitted By: | | | Score |
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# WELCOME TO OUR PROJECT REPORT

# ACKNOWLEDGEMENT

Group 6 is grateful for the opportunity to make such a productive project using our little programming knowledge. We heartedly thank our respected advisor, Rifat Ahmed Hassan (RIH), a fantastic instructor who made this course possible and more accessible for us to grasp. Our team is also grateful for having our lab instructor, Md. Abdullah Al Sayed supported us widely and helped us whenever needed. We dedicate this project to our respected advisor and instructor.

# Project

## Simple Gaming Console

**Develop a virtual gaming console that allows players to play different games, track their scores, and use various controllers.**

## Introduction

A Simple Gaming Console software program enables players to play numerous games, track and view their scores, and earn achievements. This gaming platform is beneficial as users do not have to search for other games; instead, our program has its own package of games through which users can switch to any game they want without even needing to leave the platform.

Our program enables users to feel organized and secure while saving time. Consequently, this gaming console also offers login options, a score tracker, a segment of achievements, and guidance throughout the games.

## Libraries Used

Java is an adaptable programming language with many libraries and frameworks to support various applications, such as database connectivity, networking, GUI development, and more. In our project, you will come across Java Swing and Awt Libraries, Java IO, Lang, and Util Libraries.
The following libraries were used in our program:

- java.awt
- javax.swing
- java.io
- java.util

# Project Feature Table

| Name | ID | Contribution parts: |
|---|---|---|
| Shafaat Bin Zaman | 2232015042 | Welcome<br>UserLogin<br>Options:<br>Games:<br>RockPaperScissorsGame<br>Information<br>Assembled the program and prepared the Report |
| Adrito Bhuiyan | 2232723642 | AdminLogin<br>Options:<br>Scoreboard<br>Achievements<br>Exit<br>Games:<br>NumberGuessingGame<br>Prepared Report |

# Description

## Features

### Welcome Panel

This feature mainly greets the user for using our project and includes our orientation status by showing our group number and the project name. Furthermore, there are two buttons that the user needs to click on to move on to the login panel.

### Login Panel

A login system is a critical part of applications that enable an entity to access a secure system. The user will be logged in to the system through login to manage everything. Username and Password need to be entirely exact as recorded in the system to log in successfully. Otherwise, "Incorrect Username/Password" will pop up, and the user must input again. The user will be able to manage the following parts of the system:

➔ Admin

➔ Player

### Options

The user will be able to manage the following parts of the system:

➔ Games

➔ Scoreboard

➔ Achievement

➔ Information

➔ Exit

### Games

The various games will be displayed to the user, from where the user can choose and play the game or return to the option panel. Games that are platform offers are listed below-

- Rock Paper Scissors
- Number Guessing Game

**Scoreboard**
The user can track the scores of the games, and it will display the results- total games played, how many games the user won, how many games the computer won, or how many were tied. The results are saved in a file, from which this feature reads it and displays it to the user. However, it will terminate the user from the gaming platform.

**Achievement**
The user will be able to see what challenges are up for the player and what the user can earn by playing the games. This feature will motivate and give a reason for the players to stick to the game and earn their desired achievements. It will enable the user to contrast a sense of themselves as a success.

**Information**
It displays the game manual to the user and the information on How to play the game along with the game logic.

**Exit**
A user ends their active session within the system. The logout option is a fundamental element of user authentication systems, enhancing security, protecting privacy, managing sessions efficiently, and complying with industry standards and regulations. This terminates the whole program.

# Games

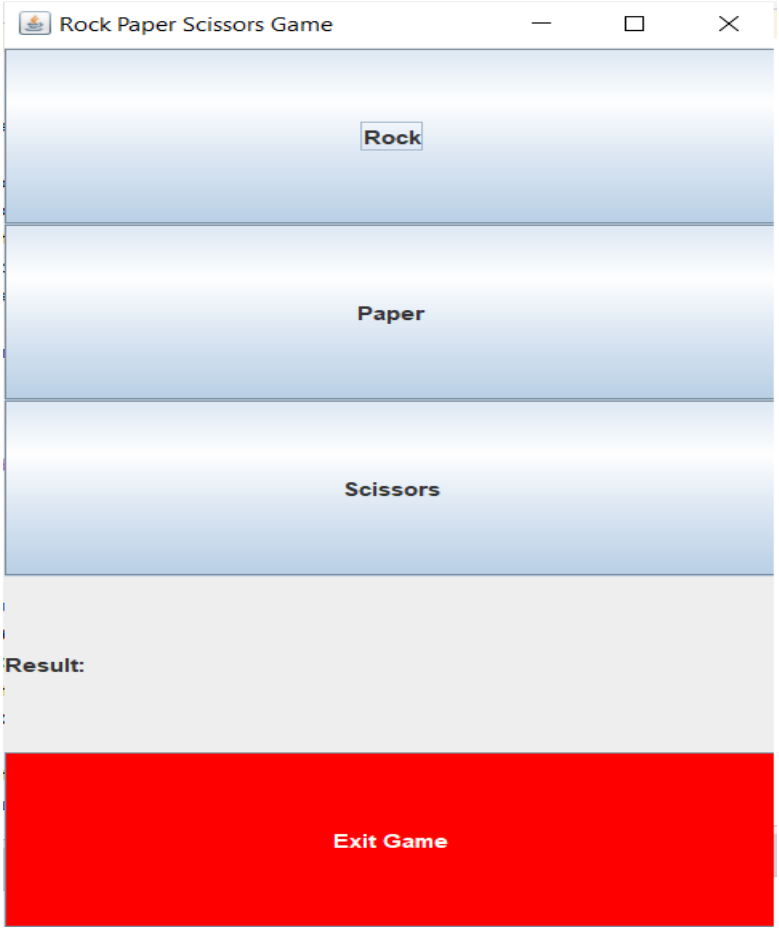
## Rock Paper Scissors Game

1. **Game Logic:** The game logic determines the winner of each round based on the choices made by the user and the computer. It follows the rules mentioned :
    - Rock crushes Scissors (Rock wins against Scissors).
    - Scissors cut Paper (Scissors win against Paper).
    - Paper covers Rock (Paper wins against Rock)
    - If both players choose the same option, the round is a tie.
    -
2. **Computer Choice:** The computer generates a random choice (Rock, Paper, or Scissors) for each round. This simulates the unpredictability of a human opponent.

3. **Score Tracking:** The game keeps track of the scores for both the user and the computer. This can be viewed from the scoreboard option in the OPTIONS panel.

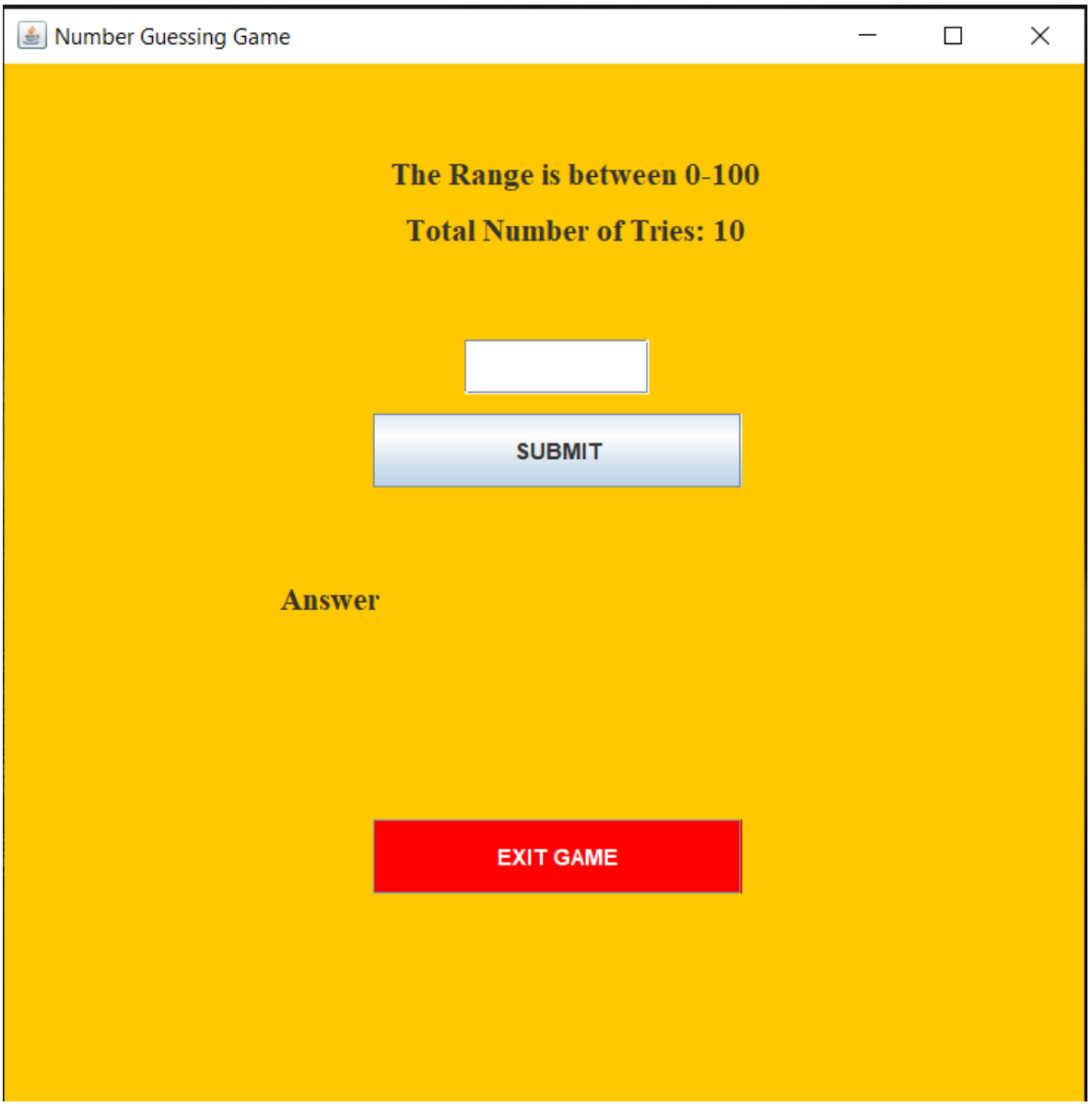
4. **Displaying Results:** The results of each round, including the choices made by the user and the computer and the winner of the game, are displayed to the user.

5. **Exit Game:** The user can keep playing other rounds or exit the game.



# Number Guessing Game

A Number Guessing Game is a simple game where the computer randomly selects a number within a specified range, and the player tries to guess that number within a limited number of guesses. The game typically provides feedback to the player after each guess, indicating whether the guessed number is too high, too low, or correct. The objective is for the player to think of the correct number with the fewest attempts possible.



## Drawbacks of the project

All of the features we intended to add, which would greatly benefit the users, still need to be displayed in our program. But we tried to complete this program to the best of our abilities.

We need help completing features, including adding other games and administrative functions, as well as creating a back option in the scoreboard.

However, our program is adaptable enough to be runnable. Indeed, we occasionally had trouble using Java Swing and Files.

## Conclusion

Our project can be upgraded in several ways, including adding new features like customizable exam templates, in-depth result analysis, permission management, and many more that we had in mind but needed more programming language skills. We have tried to create something that can be successfully and modestly used. With new updates, it can be formally utilized in many educational fields, giving us hope for a prosperous future as programmers.

# Screenshots of the Codes

## Achievements

```java
import java.awt.Container;
import java.awt.Font;
import java.awt.event.ActionEvent;
import java.awt.event.ActionListener;
import javax.swing.JButton;
import javax.swing.JFrame;
import javax.swing.JLabel;

public class Achievements {

    public static void main(String[] args) {
        Achievement tf = new Achievement();
    }
}

class Achievement extends JFrame {

    private JLabel l1, l2, l3, l4, l5, l6, l7, l8, l9;
    private JButton back;
    private Container c;
    private Font f;
    private Font g;

    public Achievement() {

        setVisible(b: true);
        setSize(width: 600, height: 600);
        setTitle(title: "Achivement");
        setLocationRelativeTo(c: null);
        setDefaultCloseOperation(operation: JFrame.EXIT_ON_CLOSE);

        c = this.getContentPane();
        c.setLayout(mgr:null);
        c.setBackground(c: Color.LIGHT_GRAY);

        f = new Font(name: "Times New Roman", style: Font.BOLD, size: 17);
        g = new Font(name: "Times New Roman", style: Font.BOLD, size: 14);

        l7 = new JLabel(text: "!! ACHIEVEMENTS YOU CAN EARN !!");
        l7.setBounds(x: 150, y: 5, width: 400, height: 100);
        l7.setForeground(fg: Color.BLUE);
        l7.setFont(font: f);

        c.add(comp: l7);

        l1 = new JLabel(text: "  \\*** Rock Paper Scissors***/ ");
        l1.setBounds(x: 0, y: 50, width: 400, height: 100);
        l1.setFont(font: f);
        c.add(comp: l1);

        l2 = new JLabel(text: "• Win 10times to achieve Master Key***>>> || || <<<***");
        l2.setBounds(x: 150, y: 100, width: 500, height: 100);
        l2.setFont(font: g);

        c.add(comp: l2);

        l3 = new JLabel(text: "• Win 20times and you will be called a Legend !!!!");
        l3.setBounds(x: 150, y: 150, width: 500, height: 100);
        l3.setFont(font: g);
        c.add(comp: l3);

        l9 = new JLabel(text: "• Win 10times in a row ,you will be one in {Million}");
        l9.setBounds(x: 150, y: 200, width: 500, height: 100);
        l9.setFont(font: g);
        c.add(comp: l9);

        l8 = new JLabel(text: " \\*** Number Guessing Game  ***/");
        l8.setBounds(x: 0, y: 250, width: 400, height: 100);
        l8.setFont(font: f);
        c.add(comp: l8);

        l4 = new JLabel(text: "• Guess the right answer in 20 try get ..Legendary Card..");
        l4.setBounds(x: 150, y: 300, width: 500, height: 100);
        l4.setFont(font: g);

        c.add(comp: l4);

        l5 = new JLabel(text: "• Guess right in the fist 7 try to be {Master}  ");
        l5.setBounds(x: 150, y: 350, width: 500, height: 100);
        l5.setFont(font: g);

        c.add(comp: l5);

        l6 = new JLabel(text: "• Guess in the first try to be the --Ultimate Champion--");
        l6.setBounds(x: 150, y: 400, width: 500, height: 100);
        l6.setFont(font: g);

        c.add(comp: l6);

        back = new JButton(text: "Back");
        back.setBounds(x: 200, y: 500, width: 200, height: 40);
        back.setForeground(fg: Color.BLACK);
        back.setBackground(bg: Color.GRAY);
        c.add(comp: back);

        back.setBackground(bg: Color.GRAY);
        c.add(comp: back);

        back.addActionListener(new ActionListener() {
            public void actionPerformed(ActionEvent ae) {
                new Options();
                dispose();
            }
        });
    }

}
```

# Login (AdminLog)

```java
package project;

import java.awt.Color;
import static java.awt.Color.black;
import static java.awt.Color.green;
import static java.awt.Color.white;
import java.awt.Container;
import java.awt.Font;
import java.awt.event.ActionEvent;
import java.awt.event.ActionListener;
import javax.swing.JButton;
import javax.swing.JFrame;
import javax.swing.JLabel;

public class Main extends JFrame {

    private Container c;
    private JLabel welcomeMessage,jl;
    private Font f1, f2,f3;
    private JButton admin, student;

    Main() {
        c = this.getContentPane();
        c.setLayout(mgr:null);
        c.setBackground(c: Color.yellow);

        f1 = new Font(name: "Times New Roman", style: Font.BOLD, size: 60);
        f2 = new Font(name: "Times New Roman", style: Font.BOLD, size: 15);
        f3 = new Font(name: "Times New Roman", style: Font.BOLD, size: 20);

        welcomeMessage = new JLabel(text: "WELCOME");
        welcomeMessage.setBounds(x: 135, y: 50, width: 600, height: 100);
```

```java
        welcomeMessage.setBounds(x: 135, y: 50, width: 600, height: 100);
        welcomeMessage.setFont(font: f1);
        c.add(comp: welcomeMessage);

        jl = new JLabel(text: "SIMPLE GAMING CONSOLE ");
        jl.setBounds(x: 150, y: 140, width: 300, height: 100);
        jl.setFont(font: f3);
        c.add(comp: jl);

        admin = new JButton(text: "ADMIN");
        admin.setBounds(x: 215, y: 250, width: 150, height: 50);
        admin.setFont(font: f2);
        admin.setBackground(bg: green);
        c.add(comp: admin);

        student = new JButton(text: "PLAYER");
        student.setBounds(x: 215, y: 320, width: 150, height: 50);
        student.setFont(font: f2);
        student.setBackground(bg: white);
        student.setForeground(fg: black);
        c.add(comp: student);

        admin.addActionListener(new ActionListener() {
            @Override
            public void actionPerformed(ActionEvent e) {
                dispose();
                AdminLogin admin = new AdminLogin();
                admin.setVisible(true);
            }
        });
        student.addActionListener(new ActionListener() {
            @Override
```

```java
        l3.setFont(font: f2);
        c.add(comp: l3);

        pf = new JPasswordField();
        pf.setBounds(x: 205, y: 250, width: 200, height:30);
        c.add(comp: pf);

        loginButton = new JButton(text: "Login");
        loginButton.setBounds(x: 245, y: 350, width: 100, height:30);
        loginButton.setFont(font: f2);
        loginButton.setBackground(bg: green);
        c.add(comp: loginButton);

        loginButton.addActionListener(new ActionListener() {

            @Override
            public void actionPerformed(ActionEvent ae) {

                String userName = t.getText();
                String pass = pf.getText();

                if (userName.equals(anObject: "Group6") && pass.equals(anObject: "admin")) {
                    JOptionPane.showMessageDialog(parentComponent: null, message: "You're successfully logged in!");
                    dispose();
                    Options ap = new Options();
                    ap.setVisible(b: true);
                } else {
                    JOptionPane.showMessageDialog(parentComponent: null, message: "Invalid username or password");
                    t.setText(t: "");
                    pf.setText(t: "");
                }
            }
        });
    }
}
```

# Login( User Login)

```java
import static java.awt.Color.green;
import static java.awt.Color.pink;
import java.awt.Container;
import java.awt.Font;
import java.awt.event.ActionEvent;
import java.awt.event.ActionListener;
import javax.swing.JButton;
import javax.swing.JFrame;
import javax.swing.JLabel;
import javax.swing.JOptionPane;
import javax.swing.JPasswordField;
import javax.swing.JTextField;

public class UserLogin extends JFrame {

    public static void main(String[] args) {
        UserLogin a = new UserLogin();

        a.setBounds(x: 600, y: 200, width: 400, height:350);
        a.setSize(width: 600,height:600);
        a.setVisible(b: true);
        a.setTitle(title: "Login Panel");
        a.setDefaultCloseOperation(operation: JFrame.EXIT_ON_CLOSE);
    }

    private Container c;
    private JLabel l1, l2, l3;
    private JTextField t;
    private JPasswordField pf;
    private JButton loginButton;
    private Font f1, f2;

    UserLogin() {

        this.setDefaultCloseOperation(operation: JFrame.EXIT_ON_CLOSE);
```


```java
        this.setDefaultCloseOperation(operation: JFrame.EXIT_ON_CLOSE);
```


```java
        this.setBounds(x: 300, y: 50, width: 600, height:600);
        this.setTitle(title: "Login Panel");

        c = this.getContentPane();
        c.setLayout(mgr:null);
        c.setBackground(c: pink);

        f1 = new Font(name: "Times New Roman", style: Font.BOLD, size: 40);
        f2 = new Font(name: "Times New Roman", style: Font.BOLD, size: 17);

        l1 = new JLabel(text: "GAMER LOGIN");
        l1.setBounds(x: 170, y: 50, width: 400, height:100);
        l1.setFont(font: f1);
        c.add(comp: l1);

        l2 = new JLabel(text: "Username:");
        l2.setBounds(x: 100, y: 200, width: 100, height:30);
        l2.setFont(font: f2);
        c.add(comp: l2);

        t = new JTextField();
        t.setBounds(x: 205, y: 200, width: 200, height:30);
        c.add(comp: t);

        l3 = new JLabel(text: "Password:");
        l3.setBounds(x: 100, y: 250, width: 100, height:30);
        l3.setFont(font: f2);
        c.add(comp: l3);

        pf = new JPasswordField();
        pf.setBounds(x: 205, y: 250, width: 200, height:30);
        c.add(comp: pf);

        loginButton = new JButton(text: "Login");
        loginButton = new JButton(text: "Login");
        loginButton.setBounds(x: 245, y: 350, width: 100, height:30);
        loginButton.setFont(font: f2);
        loginButton.setBackground(bg: green);
        c.add(comp: loginButton);

        loginButton.addActionListener(new ActionListener() {

            @Override
            public void actionPerformed(ActionEvent ae) {

                String userName = t.getText();
                String pass = pf.~~getText~~();

                if (userName.equals(anObject: "Meow") && pass.equals(anObject: "1234")) {
                    JOptionPane.showMessageDialog(parentComponent: null, message: "You're successfully logged in!");
                    dispose();
                    Options ap = new Options();
                    ap.setVisible(b: true);
                } else {
                    JOptionPane.showMessageDialog(parentComponent: null, message: "Invalid username or password");
                    t.setText(t: "");
                    pf.setText(t: "");
                }
            }

        });

    }
}
```

# Main

```java
import java.awt.Color;
import static java.awt.Color.black;
import static java.awt.Color.green;
import static java.awt.Color.white;
import java.awt.Container;
import java.awt.Font;
import java.awt.event.ActionEvent;
import java.awt.event.ActionListener;
import javax.swing.JButton;
import javax.swing.JFrame;
import javax.swing.JLabel;

public class Main extends JFrame {

    private Container c;
    private JLabel welcomeMessage,jl;
    private Font f1, f2,f3;
    private JButton admin, student;

    Main() {
        c = this.getContentPane();
        c.setLayout(mgr: null);
        c.setBackground(c: Color.yellow);

        f1 = new Font(name: "Times New Roman", style: Font.BOLD, size: 60);
        f2 = new Font(name: "Times New Roman", style: Font.BOLD, size: 15);
        f3 = new Font(name: "Times New Roman", style: Font.BOLD, size: 20);

        welcomeMessage = new JLabel(text: "WELCOME");
        welcomeMessage.setBounds(x: 135, y: 50, width: 600, height:100);
        welcomeMessage.setFont(font: f1);
        c.add(comp: welcomeMessage);

        jl = new JLabel(text: "SIMPLE GAMING CONSOLE ");
```


```java
        jl = new JLabel(text: "SIMPLE GAMING CONSOLE ");
```


```java
        jl.setBounds(x: 150, y: 140, width: 300, height:100);
        jl.setFont(font: f3);
        c.add(comp: jl);

        admin = new JButton(text: "ADMIN");
        admin.setBounds(x: 215, y: 250, width: 150, height:50);
        admin.setFont(font: f2);
        admin.setBackground(bg: green);
        c.add(comp: admin);

        student = new JButton(text: "PLAYER");
        student.setBounds(x: 215, y: 320, width: 150, height:50);
        student.setFont(font: f2);
        student.setBackground(bg: white);
        student.setForeground(fg: black);
        c.add(comp: student);

        admin.addActionListener(new ActionListener() {
            @Override
            public void actionPerformed(ActionEvent e) {
                dispose();
                AdminLogin admin = new AdminLogin();
                admin.setVisible(b: true);
            }
        });
        student.addActionListener(new ActionListener() {
            @Override
            public void actionPerformed(ActionEvent e) {
                dispose();
                UserLogin student = new UserLogin();
                student.setVisible(b: true);
            }
        });
    });
}

    public static void main(String[] args) {
        Main frame = new Main();
        frame.setVisible(b: true);
        frame.setDefaultCloseOperation(operation: JFrame.EXIT_ON_CLOSE);
        frame.setBounds(x: 300, y: 50, width: 600, height:600);
        frame.setTitle(title: "Simple Gaming Console");

    }

}
```

# Options

```java
import java.awt.Color;
import java.awt.Container;
import java.awt.Font;
import java.awt.event.ActionEvent;
import java.awt.event.ActionListener;
import javax.swing.JButton;
import javax.swing.JFrame;
import javax.swing.JLabel;

public class Options extends JFrame {

    private Container c;
    private JLabel jl;
    private JButton b1, b2, b3, b4, b5;
    private Font f;

    public Options() {

        setVisible(b: true);
        setSize(width: 600, height:600);
        setTitle(title: "Options");
        setLocationRelativeTo(c: null);
        setDefaultCloseOperation(operation: JFrame.EXIT_ON_CLOSE);

        c = this.getContentPane();
        c.setLayout(mgr:null);
        c.setBackground(c: Color.CYAN);

        f = new Font(name: "Times New Roman", style: Font.BOLD, size: 24);

        jl = new JLabel(text: "Choose an Option ");
        jl.setBounds(x: 200, y: 40, width: 200, height:100);
        jl.setFont(font: f);
        c.add(comp: jl);
        c.add(comp: jl);

        b1 = new JButton(text: "GAMES");
        b1.setBounds(x: 150, y: 180, width: 300, height: 40);
        c.add(comp: b1);

        b2 = new JButton(text: "SCOREBOARD");
        b2.setBounds(x: 150, y: 230, width: 300, height: 40);
        c.add(comp: b2);

        b3 = new JButton(text: "ACHIEVEMENT");
        b3.setBounds(x: 150, y: 280, width: 300, height: 40);
        c.add(comp: b3);

        b4 = new JButton(text: "INFO");
        b4.setBounds(x: 150, y: 330, width: 300, height: 40);
        c.add(comp: b4);

        b5 = new JButton(text: "EXIT");
        b5.setBounds(x: 150, y: 380, width: 300, height: 40);
        b5.setForeground(fg: Color.white);
        b5.setBackground(bg: Color.red);
        c.add(comp: b5);

        b1.addActionListener(new ActionListener() {

            public void actionPerformed(ActionEvent ae) {
                Games gms = new Games();
                gms.setVisible(b: true);
                dispose();
            }
        }

    });

    b2.addActionListener(new ActionListener() {

        public void actionPerformed(ActionEvent ae) {
            Scoreboard sb = new Scoreboard();
            sb.setVisible(b: true);
            dispose();
        }
    });

    b3.addActionListener(new ActionListener() {

        public void actionPerformed(ActionEvent ae) {
            Achievement ac = new Achievement();
            ac.setVisible(b: true);
            dispose();
        }
    });

    b4.addActionListener(new ActionListener() {

        public void actionPerformed(ActionEvent ae) {
            Information i = new Information();
            i.setVisible(b: true);
    dispose();
        }

    });

    b5.addActionListener(new ActionListener() {

        public void actionPerformed(ActionEvent ae) {
            new Login();
            dispose();
        }
    });
}

public static void main(String[] args) {
    Options sp = new Options();
}
}
```

# Games

```java
import java.awt.Color;
import java.awt.Container;
import java.awt.Font;
import java.awt.event.ActionEvent;
import java.awt.event.ActionListener;
import javax.swing.JButton;
import javax.swing.JFrame;
import javax.swing.JLabel;

public class Games extends JFrame {

    public Container c;
    public JLabel jl;
    public JButton b1, b2, b3;
    public Font f;

    public Games() {

        setVisible(b: true);
        setSize(width: 500, height: 400);
        setTitle(title: "GAMES");
        setLocationRelativeTo(c: null);
        setDefaultCloseOperation(operation: JFrame.EXIT_ON_CLOSE);

        c = this.getContentPane();
        c.setLayout(mgr:null);
        c.setBackground(c: Color.DARK_GRAY);

        f = new Font(name: "Times New Roman", style: Font.BOLD, size: 20);

        jl = new JLabel(text: "CHOOSE A GAME ");
        jl.setForeground(fg: Color.WHITE);
```

```java
        jl.setForeground(fg: Color.WHITE);
        jl.setBounds(x: 170, y: 40, width: 200, height:100);
        jl.setFont(font: f);
        c.add(comp: jl);

        b1 = new JButton(text: "ROCK PAPER SCISSORS");
        b1.setBounds(x: 100, y: 140, width: 300, height:40);
        b1.setForeground(fg: Color.BLACK);
        b1.setBackground(bg: Color.white);
        c.add(comp: b1);

        b2 = new JButton(text: "NUMBER GUESSING GAME");
        b2.setBounds(x: 100, y: 190, width: 300, height:40);
        b2.setForeground(fg: Color.BLACK);
        b2.setBackground(bg: Color.white);
        c.add(comp: b2);

        b3 = new JButton(text: "BACK");
        b3.setBounds(x: 100, y: 240, width: 300, height:40);
        b3.setForeground(fg: Color.BLACK);
        b3.setBackground(bg: Color.LIGHT_GRAY);
        c.add(comp: b3);

        b1.addActionListener(new ActionListener() {

            public void actionPerformed(ActionEvent ae) {
                RockPaperScissorsGame rps = new RockPaperScissorsGame();
                rps.setVisible(b: true);
                dispose();
            }

        });

        b2.addActionListener(new ActionListener() {
```

```java
        b2.addActionListener(new ActionListener() {

            public void actionPerformed(ActionEvent ae) {
                GuessNumber gm = new GuessNumber();
                gm.setVisible(b: true);
                dispose();
            }
        });

        b3.addActionListener(new ActionListener() {
            public void actionPerformed(ActionEvent ae) {
                new Options();
                dispose();
            }
        });
    }

    public static void main(String[] args) {
        Games gms = new Games();
    }
}
```

# Scoreboard

```java
import java.awt.BorderLayout;
import javax.swing.*;
import java.io.BufferedReader;
import java.io.FileReader;
import java.io.IOException;

public class Scoreboard extends JFrame {

    private JTextArea resultsArea;
    private JButton backButton;

    public Scoreboard() {

        setTitle("SCOREBOARD");
        setSize(300, 500);
        setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        setLocationRelativeTo(null);

        resultsArea = new JTextArea();
        resultsArea.setEditable(false);

        add(new JScrollPane(resultsArea), BorderLayout.CENTER);

        loadResultsFromFile();

    }

    private void loadResultsFromFile() {
        try (BufferedReader reader = new BufferedReader(new FileReader("game_results.txt"))) {
            String line;
            while ((line = reader.readLine()) != null) {
                resultsArea.append(line + "\n");
```

```java
    }

    private void loadResultsFromFile() {
        try (BufferedReader reader = new BufferedReader(new FileReader("game_results.txt"))) {
            String line;
            while ((line = reader.readLine()) != null) {
                resultsArea.append(line + "\n");
            }
        } catch (IOException e) {
            e.printStackTrace();
            resultsArea.append("Error reading results from file.");
        }
    }

    public static void main(String[] args) {

        Scoreboard view = new Scoreboard();
        view.setVisible(true);

    }
}
```

# INFO

```java
import java.awt.Color;
import java.awt.Container;
import java.awt.Font;
import java.awt.event.ActionEvent;
import java.awt.event.ActionListener;
import javax.swing.JButton;
import javax.swing.JFrame;
import javax.swing.JLabel;

public class Info {

    public static void main(String[] args) {
        Information tf = new Information();
    }
}

class Information extends JFrame {

    private JLabel l1, l2, l3, l4, l5, l6, l7, l8,l9;
    private JButton back;
    private Container c;
    private Font f,g,h;

    public Information() {

        setVisible(b: true);
        setSize(width: 600, height: 600);
        setTitle(title: "How to Play");
        setLocationRelativeTo(c: null);
        setDefaultCloseOperation(operation: JFrame.EXIT_ON_CLOSE);

        c = this.getContentPane();
        c.setLayout(mgr:null);
        c.setBackground(c: Color.LIGHT_GRAY);

        f = new Font(name: "Times New Roman", style: Font.ITALIC, size: 19);
        g = new Font(name: "Times New Roman", style: Font.PLAIN, size: 14);
        h = new Font(name: "Times New Roman", style: Font.BOLD, size: 22);

        l7 = new JLabel(text: "HOW TO PLAY");
        l7.setBounds(x: 220, y: 0, width: 400, height: 100);
        l7.setFont(font: h);
        l7.setForeground(fg: Color.blue);
        c.add(comp: l7);

        l1 = new JLabel(text: "  \\*** ROCK PAPER SCISSORS ***//");
        l1.setBounds(x: 150, y: 40, width: 400, height: 100);
        l1.setFont(font: f);
        c.add(comp: l1);

        l2 = new JLabel(text: "@ Logics -  ");
        l2.setBounds(x: 0, y: 80, width: 500, height: 100);
        l2.setFont(font: g);
        c.add(comp: l2);

        l9 = new JLabel(text: "Rock beats Scissors , Scissors beats Paper , Paper beats Rock  ");
        l9.setBounds(x: 0, y: 120, width: 500, height: 100);
        l9.setFont(font: g);
        c.add(comp: l9);

        l3 = new JLabel(text: "When a button is clicked, the game logic is executed and the result is displayed ");
        l3.setBounds(x: 0, y: 160, width: 500, height: 100);
        l3.setFont(font: g);
        l3.setFont(font: g);
        c.add(comp: l3);

        l8 = new JLabel(text: " \\*** NUMBER GUESSING GAME  ***//");
        l8.setBounds(x: 150, y: 210, width: 400, height: 100);
        l8.setFont(font: f);
        c.add(comp: l8);

        l4 = new JLabel(text: "@ Logics - .");
        l4.setBounds(x: 0, y: 240, width: 500, height: 100);
        l4.setFont(font: g);
        c.add(comp: l4);

        l5 = new JLabel(text: " The Game generates a random number between 1 - 100");
        l5.setBounds(x: 0, y: 280, width: 500, height: 100);
        l5.setFont(font: g);
        c.add(comp: l5);

        l6 = new JLabel(text: "Player has to guess the exact correct number");
        l6.setBounds(x: 0, y: 320, width: 450, height: 100);
        l6.setFont(font: g);
        c.add(comp: l6);

        back = new JButton(text: "Back");
        back.setBounds(x: 200, y: 500, width: 200, height: 40);
        back.setForeground(fg: Color.BLACK);
        back.setBackground(bg: Color.GRAY);
        c.add(comp: back);

        back.addActionListener(new ActionListener() {
            public void actionPerformed(ActionEvent ae) {
```

```java
        l6.setBounds(x: 0, y: 320, width: 450, height: 100);
        l6.setFont(font: g);
        c.add(comp: l6);

        back = new JButton(text: "Back");
        back.setBounds(x: 200, y: 500, width: 200, height: 40);
        back.setForeground(fg: Color.BLACK);
        back.setBackground(bg: Color.GRAY);
        c.add(comp: back);

        back.addActionListener(new ActionListener() {
            public void actionPerformed(ActionEvent ae) {
                new Options();
                dispose();
            }
        });
    }

    }

}
```

# Rock Paper ScissorGame

```java
import java.awt.Color;
import java.awt.GridLayout;
import java.awt.event.ActionEvent;
import java.awt.event.ActionListener;
import java.io.BufferedWriter;
import java.io.FileWriter;
import java.io.IOException;
import javax.swing.JButton;
import javax.swing.JFrame;
import javax.swing.JLabel;
import javax.swing.JOptionPane;

public class RockPaperScissorsGame extends JFrame implements ActionListener {

    public JButton rockButton, paperButton, scissorsButton,exitButton;
    public JLabel resultLabel;
    public int totalGames = 0;
    public int playerWins = 0;
    public int computerWins = 0;
    public int tiedGames = 0;

    public RockPaperScissorsGame() {
        setTitle("Rock Paper Scissors Game");
        setSize(400, 600);
        setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        setLocationRelativeTo(null);

        rockButton = new JButton("Rock");
        paperButton = new JButton("Paper");
        scissorsButton = new JButton("Scissors");
        exitButton = new JButton("Exit Game");
        exitButton.setForeground(Color.white);
        exitButton.setForeground(Color.white);
        exitButton.setBackground(Color.red);


        exitButton.addActionListener(new ActionListener() {

            public void actionPerformed(ActionEvent ae) {
                Games gms = new Games();
                gms.setVisible(true);
                dispose();
            }
        });


        resultLabel = new JLabel("Result: ");

        rockButton.addActionListener(this);
        paperButton.addActionListener(this);
        scissorsButton.addActionListener(this);

        setLayout(new GridLayout(rows: 5, cols: 1));

        add(comp: rockButton);
        add(comp: paperButton);
        add(comp: scissorsButton);
        add(comp: resultLabel);
        add(comp: exitButton);
    }

    @Override
    public void actionPerformed(ActionEvent e) {
        String playerChoice = "";
```

```java
    public void actionPerformed(ActionEvent e) {
        String playerChoice = "";
        String computerChoice = "";

        if (e.getSource() == rockButton) {
            playerChoice = "Rock";
        } else if (e.getSource() == paperButton) {
            playerChoice = "Paper";
        } else if (e.getSource() == scissorsButton) {
            playerChoice = "Scissors";
        }

        String[] choices = {"Rock", "Paper", "Scissors"};
        int computerChoiceIndex = (int) (Math.random() * choices.length);
        computerChoice = choices[computerChoiceIndex];

        String result = determineWinner(playerChoice, computerChoice);

        resultLabel.setText("Result: " + result);

        updateCounts(result);
        achievementPop();

        saveResultToFile(result);
    }

    public String determineWinner(String playerChoice, String computerChoice) {
        totalGames++;
        if (playerChoice.equals(anObject: computerChoice)) {
            return "It's a tie!";
        } else if ((playerChoice.equals(anObject: "Rock") && computerChoice.equals(anObject: "Scissors")) ||
                (playerChoice.equals(anObject: "Paper") && computerChoice.equals(anObject: "Rock")) ||
                (playerChoice.equals(anObject: "Scissors") && computerChoice.equals(anObject: "Paper"))) {
```

```java
                (playerChoice.equals(anObject: "Scissors") && computerChoice.equals(anObject: "Paper"))) {
            return "You win!";
        } else {
            return "Computer wins!";
        }
    }

    public void updateCounts(String result) {
        if (result.equals(anObject: "It's a tie!")) {
            tiedGames++;
        } else if (result.equals(anObject: "You win!")) {
            playerWins++;
        } else if (result.equals(anObject: "Computer wins!")) {
            computerWins++;
        }
    }

    public void saveResultToFile(String result) {
        try (BufferedWriter writer = new BufferedWriter(new FileWriter(fileName: "game_results.txt", append:true))) {

            writer.write("Total Games Played: " + totalGames);
            writer.newLine();
            writer.write("Result: " + result);
            writer.newLine();
            writer.write("User Wins: " + playerWins);
            writer.newLine();
            writer.write("Computer Wins: " + computerWins);
            writer.newLine();
            writer.write("Tied Games: " + tiedGames);
            writer.newLine();
            writer.newLine();
        } catch (IOException e) {
            e.printStackTrace();
```


```java
            writer.newLine();
        } catch (IOException e) {
            e.printStackTrace();
```


```java
        }
    }

    public void achievementPop() {
        if(playerWins == 10) {
            JOptionPane.showMessageDialog(parentComponent: null, message: "!!Congrats!! You earned Master Key");
            RockPaperScissorsGame rps = new RockPaperScissorsGame();
                rps.setVisible(b: true);
                dispose();
        } else if(playerWins == 20) {
            JOptionPane.showMessageDialog(parentComponent: null, message: "!!Congrats!! You earned the Legend Title");
            RockPaperScissorsGame rps = new RockPaperScissorsGame();
                rps.setVisible(b: true);
                dispose();
        }
    }

    public static void main(String[] args) {
        RockPaperScissorsGame game = new RockPaperScissorsGame();
        game.setVisible(b: true);
    }
}
```

# NumberGuessing Game

```java
import java.awt.Color;
import java.awt.Container;
import java.awt.Font;
import java.awt.event.ActionEvent;
import java.awt.event.ActionListener;
import javax.swing.JButton;
import javax.swing.JFrame;
import javax.swing.JLabel;
import javax.swing.JOptionPane;
import javax.swing.JTextField;

public class NumberGuessingGame {

    public static void main(String[] args) {
        GuessNumber guess = new GuessNumber();
    }
}

class GuessNumber extends JFrame {

    private JTextField t1;
    private JLabel l1, l2,l4;
    private JButton b1, b2;
    private Container c;
    private Font f;

    int count = 0;
    int gg = 10;
    int high = 0;
    int low = 0;

    int m = (int) (Math.random() * 100);
    //int m = 5;

    public GuessNumber() {

        setVisible(b: true);
        setSize(width: 600, height: 600);
        setTitle(title: "Number Guessing Game");
        setLocationRelativeTo(c: null);
        setDefaultCloseOperation(operation: JFrame.EXIT_ON_CLOSE);

        c = this.getContentPane();
        c.setLayout(mgr:null);
        c.setBackground(c: Color.orange);

        f = new Font(name: "Times New Roman", style: Font.BOLD, size: 17);

        l1 = new JLabel(text: "The Range is between 0-100 ");
        l1.setBounds(x: 210, y: 10, width: 600, height: 100);
        l1.setFont(font: f);
        c.add(comp: l1);

        l4 = new JLabel(text: "  Total Number of Tries: 10");
        l4.setBounds(x: 210, y: 80, width: 600, height: 20);
        l4.setFont(font: f);
        c.add(comp: l4);


        t1 = new JTextField(columns:20);
        t1.setBounds(x: 250, y: 150, width: 100, height: 30);
        c.add(comp: t1);

        b1 = new JButton(text: "SUBMIT");
        b1.setBounds(x: 200, y: 190, width: 200, height: 40);
        c.add(comp: b1);

        l2 = new JLabel(text: "Answer");
        l2.setBounds(x: 150, y: 240, width: 600, height: 100);
        l2.setFont(font: f);
        c.add(comp: l2);

        b2 = new JButton(text: "EXIT GAME");
        b2.setBounds(x: 200, y: 410, width: 200, height: 40);
        b2.setForeground(fg: Color.white);
        b2.setBackground(bg: Color.red);
        c.add(comp: b2);

        b1.addActionListener(new ActionListener() {
            public void actionPerformed(ActionEvent ae) {
                int num = Integer.parseInt(s: t1.getText());

                if ((num == m && gg > 0) && (num == m && gg <5)) {
                    gg--;
                    JOptionPane.showMessageDialog(parentComponent:null, message:"Hurray !! You guessed the right number");
                    GuessNumber gm = new GuessNumber();
                    gm.setVisible(b: true);
                    dispose();

                } else if((num == m && gg > 4) && (num == m && gg <7)){
                    gg--;
                    JOptionPane.showMessageDialog(parentComponent:null, message:"!!Congrats!! You earned Rookie TITLE !");
```


```java
                    gg--;
                    JOptionPane.showMessageDialog(parentComponent:null, message:"!!Congrats!! You earned Rookie TITLE !");
```


```java
                    GuessNumber gm = new GuessNumber();
                    gm.setVisible(b: true);
                    dispose();

                } else if((num == m && gg > 6) && (num == m && gg <9)){
                    gg--;
                    JOptionPane.showMessageDialog(parentComponent:null, message:"!!Congrats!! You earned MASTER GUARDIAN TITLE !");
                    GuessNumber gm = new GuessNumber();
                    gm.setVisible(b: true);
                    dispose();

                } else if(num == m & gg>=9) {
                    gg--;
                    JOptionPane.showMessageDialog(parentComponent:null, message:"!!HATS OFF !! YOU are the ULTIMATE CHAMPION");
                    GuessNumber gm = new GuessNumber();
                    gm.setVisible(b: true);
                    dispose();

                } else if (num < m && gg > 1 && num<100) {
                    //high++;
                    gg--;
                    l2.setText("Oops..Guess Higher!"+" You have "+gg+" tries left");

                } else if (num > m && gg > 1 && num <100) {
                    // low++;
                    gg--;
                    l2.setText("Oops..Guess Lower!"+" You have "+gg+" tries left.");

                } else if (num >= 100) {
                    l2.setText("Please Guess Between 0 to 100!"+" You still have "+gg+" tries left.");
```


```java
                } else if (num >= 100) {
                    l2.setText("Please Guess Between 0 to 100!"+" You still have "+gg+" tries left.");
```


```java
                } else {
                    JOptionPane.showMessageDialog(parentComponent:null, message:"Sorry You lost! Please Try again");
                    GuessNumber gm = new GuessNumber();
                    gm.setVisible(b: true);
                    dispose();

                }
            }
        });

        b2.addActionListener(new ActionListener() {

            public void actionPerformed(ActionEvent ae) {
                Games gms = new Games();
                gms.setVisible(b: true);
                dispose();
            }
        });


    }

}
```

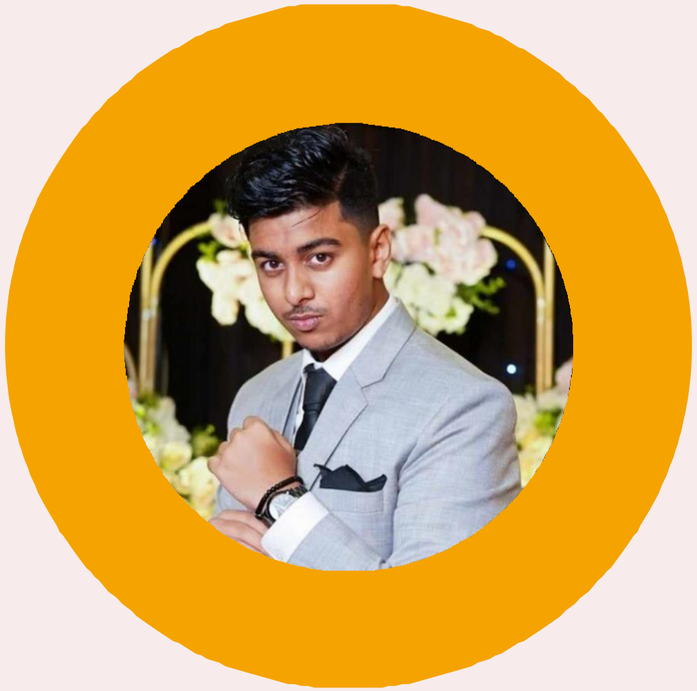
# Simple Gaming Console ( Welcome )

```java
import java.awt.Color;
import java.awt.Container;
import javax.swing.JFrame;
import javax.swing.JLabel;
import java.awt.Font;
import java.awt.event.ActionEvent;
import java.awt.event.ActionListener;
import javax.swing.JButton;

public class SimpleGamingConsole extends JFrame {

    private Container c;
    private JLabel l1, l2;
    private JButton b1;
    private Font f;

    public SimpleGamingConsole() {

        c = this.getContentPane();
        c.setLayout(mgr: null);
        c.setBackground(c: Color.YELLOW);

        f = new Font(name: "Times New Roman", style: Font.BOLD, size: 17);

        l1 = new JLabel(text: "WELCOME TO GROUP-6's PROJECT");
        l1.setBounds(x: 150, y: 100, width: 300, height: 200);
        l1.setFont(font: f);
        c.add(comp: l1);

        l2 = new JLabel(text: "SIMPLE GAMING CONSOLE");
```


```java
        l2 = new JLabel(text: "SIMPLE GAMING CONSOLE");
```


```java
        l2.setBounds(x: 177, y: 230, width: 300, height: 30);
        l2.setFont(font: f);
        c.add(comp: l2);

        b1 = new JButton(text: "ENTER");
        b1.setBounds(x: 238, y: 270, width: 100, height: 30);
        c.add(comp: b1);

        b1.addActionListener(new ActionListener() {

            public void actionPerformed(ActionEvent e) {
                Options os = new Options();
                    os.setVisible(b: true);
                    dispose();

            }
        });

        setVisible(b: true);
        setSize(width: 600, height: 600);
        setLocationRelativeTo(c: null);
        setDefaultCloseOperation(operation: JFrame.EXIT_ON_CLOSE);

    }

    public static void main(String[] args) {

        SimpleGamingConsole sgc = new SimpleGamingConsole();
        sgc.setVisible(b: true);

    }
```

# TEAM MEMBERS



**SHAFAAT BIN ZAMAN**

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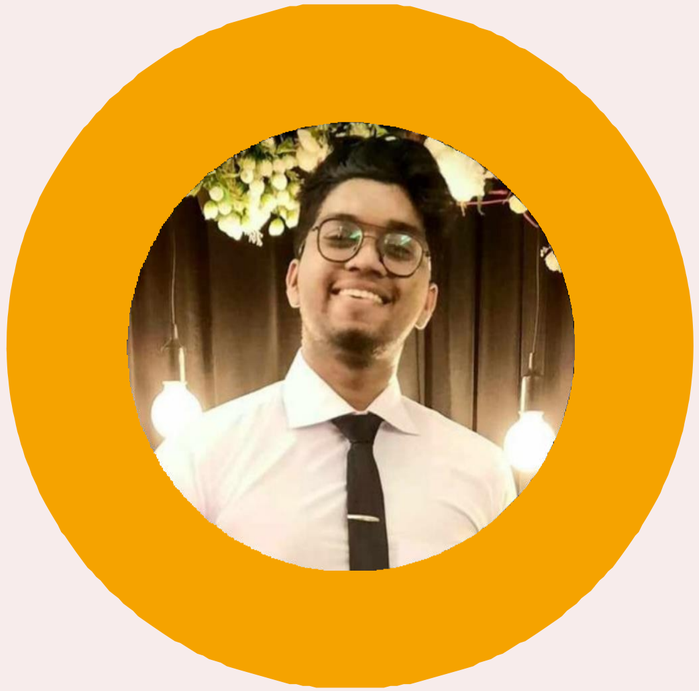

**ADRITO BHUIYAN**

2232723642

# THANK YOU!